# Neural Network Field Guide

## 42 architectures and training schemes, from the perceptron (1958) to diffusion LLMs (2025).

Each diagram draws a network in miniature, input at the bottom and output at the top; one that settles as a whole is a ring, and a graph network is its graph. Most circles are neurons; in sequence models a column is a token, a circle its vector at one layer. A square is larger: an image patch, a matrix-valued state, a whole network, or an environment. Color and mark both show a cell's role, so diagrams read without color. Faded parts play little role in the step shown. Parts repeated over positions, windows, copies, or passes share weights; layers, experts, encoders, and decoders drawn alike do not. Sections group by trait: wiring, dynamics, training, cost, or purpose. Years and methods follow landmark papers, not always the first.

**Input**
- Input unit
- Image patch
- Noisy input
- Random noise
- Visible unit (in and out)
- Masked token

**Hidden**
- Hidden unit
- Probabilistic hidden
- Capsule (vector output)
- Router

**State**
- Recurrent unit
- Memory cell
- Matrix-valued state

**Mixing**
- Attention
- Convolution or pooling

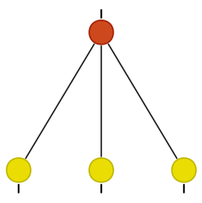
**Output**
- Output unit
- Reconstruction of input
- Velocity

**Whole systems**
- A whole network
- Environment

**Links**
- Connection
- Skip connection
- Data passed on or fed back
- Gate, acting at the dot
- Similarity score
- Learnable function

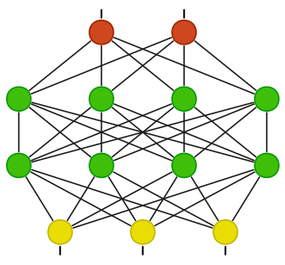
### FEED-FORWARD

#### Perceptron



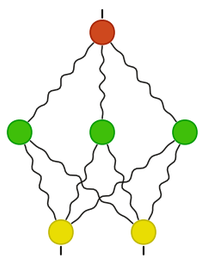
Weighted sum, then a threshold
`1958 · supervised`

#### MLP



Stacked dense layers
`1986 · supervised`

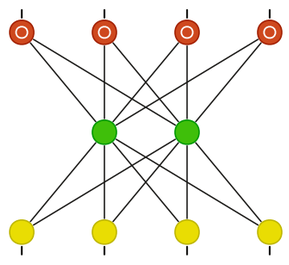
#### KAN



Learnable functions on edges
`2024 · supervised`

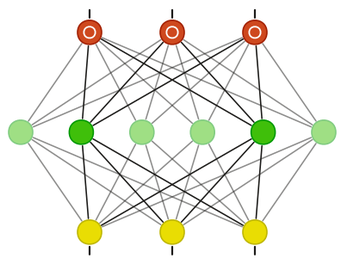
### AUTOENCODERS

#### Autoencoder



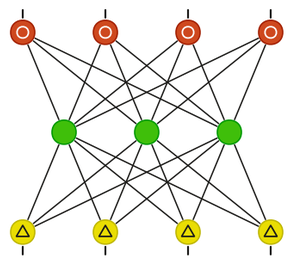
Encode, then reconstruct
`1986 · reconstruct`

#### Sparse autoencoder



Few active units per input
`2006 · reconstruct`

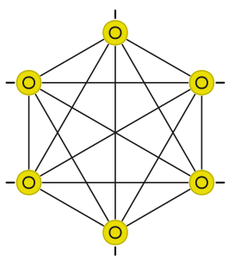
#### Denoising autoencoder



Clean up a noisy input
`2008 · denoise`

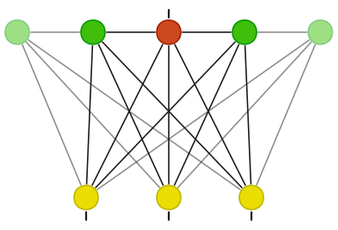
### ENERGY-BASED AND SELF-ORGANIZING

#### Hopfield network



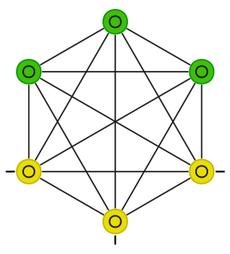
Settles into an energy minimum
`1982 · hebbian`

#### Kohonen map



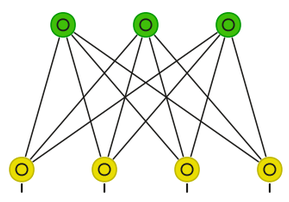
Winner and neighbors move
`1982 · competitive`

#### Boltzmann machine



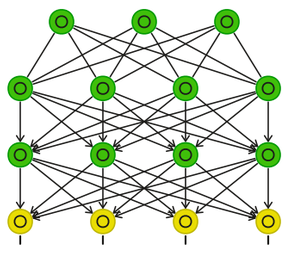
Stochastic units, fully connected
`1985 · likelihood`

#### RBM



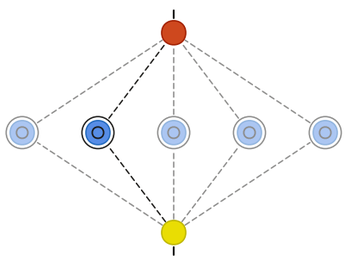
No links within a layer
`1986 · likelihood`

#### Deep belief network



Stacked RBMs, trained greedily
`2006 · likelihood`

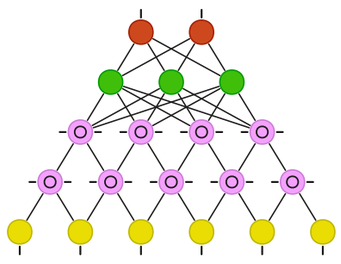
#### Modern Hopfield network



Attention as memory lookup
`2020 · supervised`

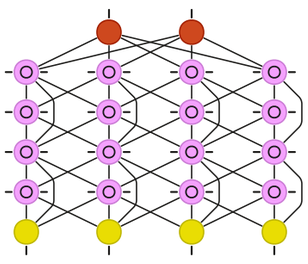
### CONVOLUTIONAL

#### CNN



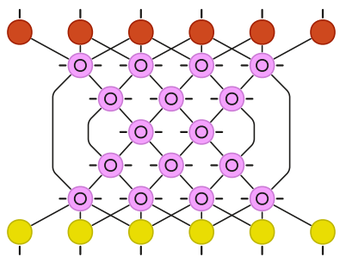
Local filters, shared weights
`1989 · supervised`

#### ResNet



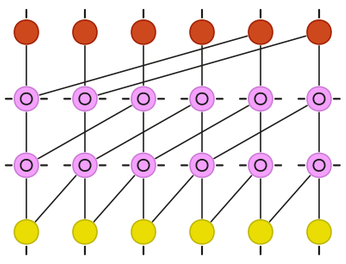
Blocks add to a skip path
`2015 · supervised`

#### U-Net



Shrink, expand, skip across
`2015 · supervised`

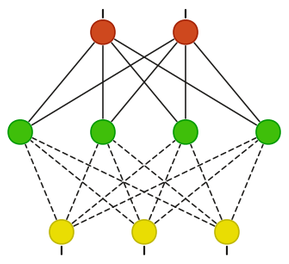
#### WaveNet



Dilated causal convolutions
`2016 · next token`

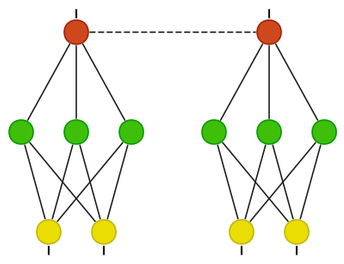
### SIMILARITY

#### Radial basis network



Distance to stored centers
`1988 · supervised`

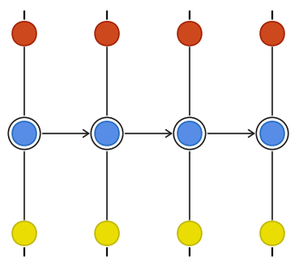
#### Siamese network



Twin towers, shared weights
`1993 · contrastive`

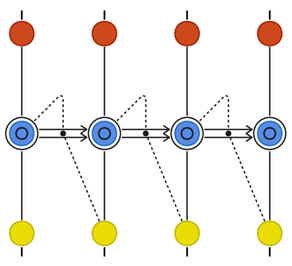
### RECURRENT AND MEMORY

#### RNN



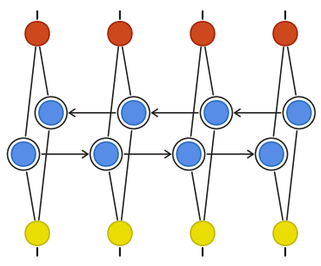
Hidden state passed along
`1990 · next token`

#### LSTM



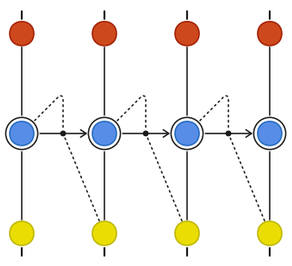
Two states; forget gate from 1999
`1997 · next token`

#### Bidirectional RNN



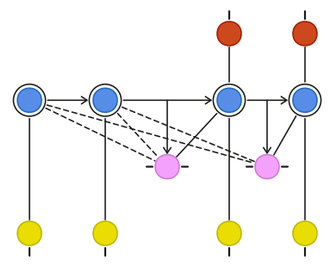
Reads the sequence both ways
`1997 · supervised`

#### GRU



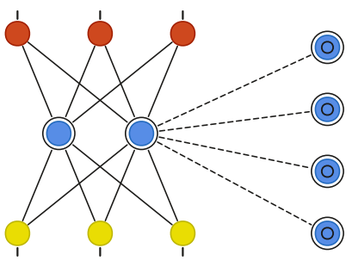
One state, two gates
`2014 · next token`

#### Seq2seq with attention



Learns to align and translate
`2014 · next token`

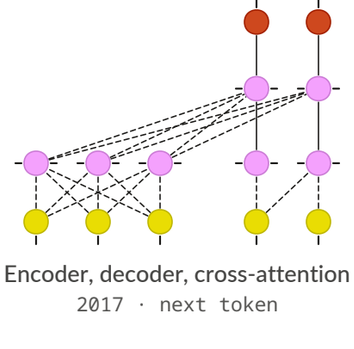
#### Neural Turing machine



Controller reads, writes memory
`2014 · supervised`

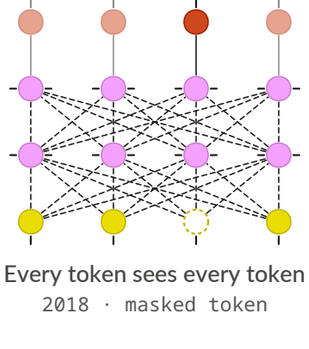
### ATTENTION

#### Transformer



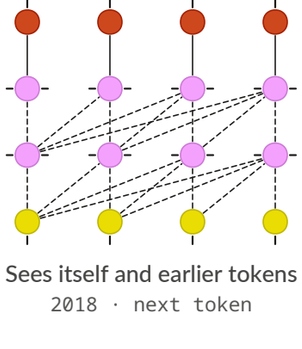
Encoder, decoder, cross-attention
`2017 · next token`

#### BERT



Every token sees every token
`2018 · masked token`

#### GPT-style LLM



Sees itself and earlier tokens
`2018 · next token`

#### ViT

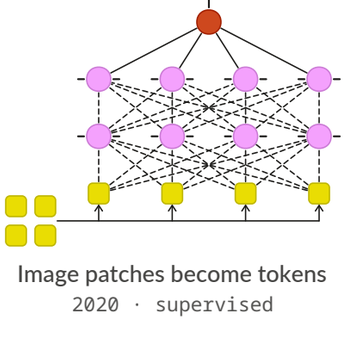
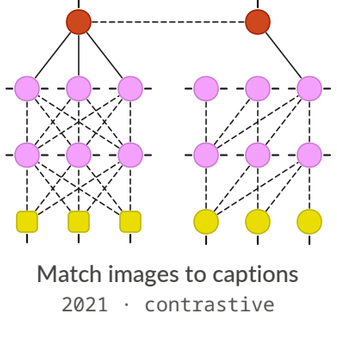


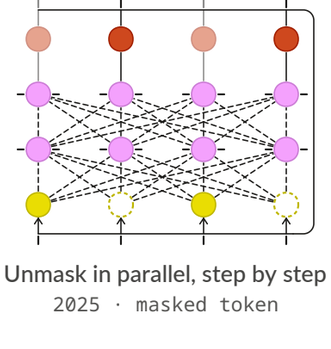
Image patches become tokens
`2020 · supervised`

#### CLIP



Match images to captions
`2021 · contrastive`

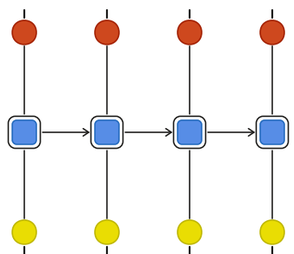
#### Masked diffusion LLM



Unmask in parallel, step by step
`2025 · masked token`

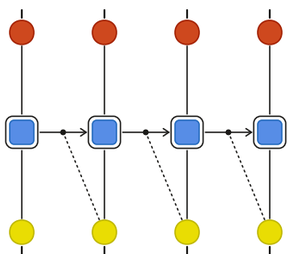
### LINEAR-TIME MODELS AND HYBRIDS

#### Linear attention



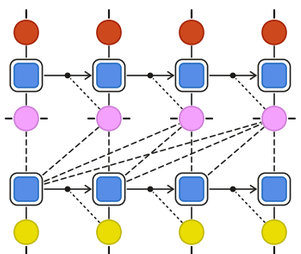
Softmax-free attention as an RNN
`2020 · next token`

#### Mamba



Selective state, linear time
`2023 · next token`

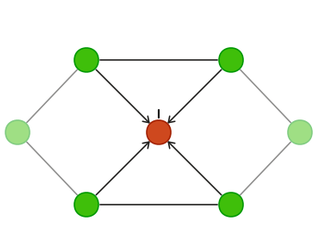
#### Recurrent hybrid



Mostly recurrent, some attention
`2024 · next token`

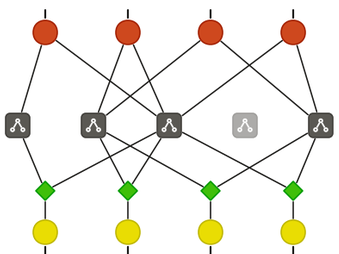
### ROUTING AND GRAPHS

#### Graph neural network



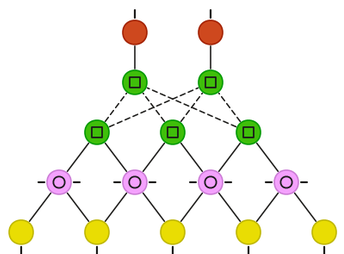
Nodes pool neighbor messages
`2005 · supervised`

#### Sparse MoE



Each token → its top-*k* experts
`2017 · next token`

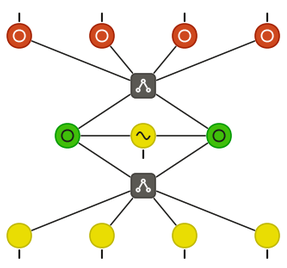
#### Capsule network



Vectors routed by agreement
`2017 · supervised`

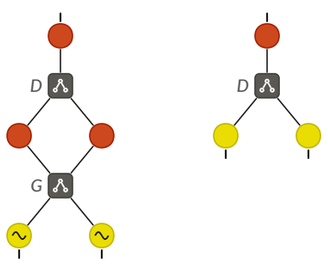
### GENERATIVE FRAMEWORKS

#### Variational autoencoder



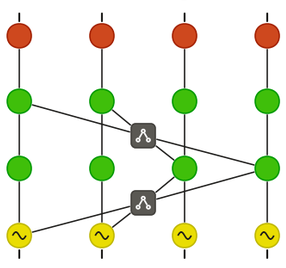
Sample a code, then decode
`2013 · variational`

#### GAN



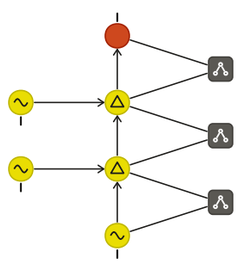

Generator vs. discriminator
`2014 · adversarial`

#### Normalizing flow



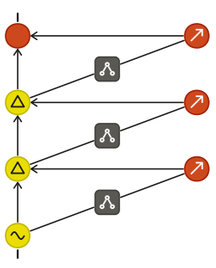
Invertible map from noise to data
`2014 · likelihood`

#### Diffusion model



One denoiser, run *T* times
`2015 · variational`

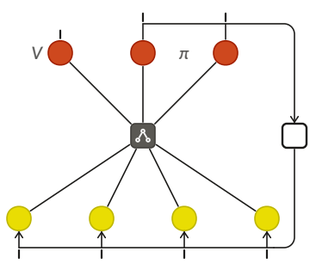
#### Flow matching



Learns a velocity, noise to data
`2022 · flow matching`

### REINFORCEMENT

#### Actor-critic (PPO)




More of what earned reward
`2017 · reward`

Inspired by The Neural Network Zoo by Fjodor van Veen (Asimov Institute, 2016; updated with Stefan Leijnen, 2019). www.asimovinstitute.org/neural-network-zoo

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